

REPORT

Title

Implementation of Depth First Search (DFS) Algorithm Using Python

Aim

To implement the Depth First Search (DFS) algorithm in Python for traversing a graph from a specified starting vertex.

Problem Statement

Depth First Search (DFS) is a graph traversal algorithm used to visit all the vertices of a graph by exploring one branch completely before moving to another. The algorithm starts from a given source vertex and continues visiting adjacent unvisited vertices until it reaches a dead end, after which it backtracks to explore the remaining vertices. DFS can operate on any graph represented using an adjacency list or adjacency matrix and is applicable to both directed and undirected graphs.

Requirements

- Python 3.x

Procedure

1. Read the number of vertices from the user.
2. Accept the graph as an adjacency list.

3. Read the starting vertex.
4. Initialize an empty set to keep track of visited vertices.
5. Visit the starting vertex and mark it as visited.
6. Recursively visit each unvisited adjacent vertex.
7. Backtrack when a vertex has no unvisited neighbors.
8. Continue until all reachable vertices have been visited.
9. Display the traversal order.

Algorithm

1. Start from the source vertex.
2. Mark the current vertex as visited.
3. Display the current vertex.
4. Traverse each adjacent vertex.
5. If an adjacent vertex has not been visited, recursively perform DFS on it.
6. Repeat until all reachable vertices are visited.

Python Program

```
graph = {}

n = int(input("Enter the number of vertices: "))

for i in range(n):
    vertex = input("Enter vertex: ")
    neighbors = input(f"Enter adjacent vertices of {vertex} (space-separated): ").split()
```

```
graph[vertex] = neighbors

start = input("Enter the starting vertex: ")

visited = set()

def dfs(vertex):
    if vertex not in visited:
        print(vertex, end=" ")
        visited.add(vertex)

        for neighbor in graph.get(vertex, []):
            dfs(neighbor)

print("DFS Traversal:")
dfs(start)
```

Sample Output

```
DFS Traversal:
A B D E F C
```

Result

The Depth First Search algorithm was successfully implemented in Python. The program correctly traverses any user-defined graph from the specified

starting vertex by exploring one path completely before backtracking and visiting the remaining vertices.

Conclusion

Depth First Search is an efficient graph traversal technique that systematically explores all reachable vertices from a starting node. It is widely used in artificial intelligence, graph theory, path finding, cycle detection, topological sorting, connected component analysis, and maze-solving problems due to its recursive and backtracking nature.